■ B.3.6 Level Specifications

n	levels `1` through n
`Infinity`	all levels
`{`n`}`	level n only
`{`n_1`, `n_2`}`	levels n_1 through n_2

Level specifications.

The level in an expression corresponding to a non-negative integer n is defined to consist of parts specified by n non-negative indices. A negative level number `-`n represents all parts of an expression that have depth n. The depth of an expression, `Depth[`*expr*`]`, is the maximum number of indices needed to specify any part, plus one. Levels *do not* include heads of expressions. Level `0` is the whole expression. Level `-1` contains all symbols and other objects that have no subparts.

Ranges of levels specified by `{`n_1`, `n_2`}` contain all parts that are neither above level n_1, nor below level n_2 in the tree. The n_i need not have the same sign. Thus, for example, `{2, -2}` specifies subexpressions which occur anywhere below the top level, but above the leaves, of the expression tree.

Level specifications are used by the functions `Apply`, `Count`, `Level`, `Map` and `Scan`.

Functions with level specifications visit different subexpressions in an order that corresponds to depth-first traversal of the expression tree, with leaves visited before roots. The subexpressions visited have part specifications which occur in an order which is lexicographic, except that longer sequences appear before shorter ones.